

From rational linear programming to positive linear programming: an exact sparse reduction for threshold decisions

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Abstract

We give a deterministic polynomial-time reduction from the threshold-decision problem for binary-encoded rational linear programs to an exact positive packing-LP threshold problem. Given $A \in \mathbb{Q}^{m \times n}$, $b \in \mathbb{Q}^m$, $c \in \mathbb{Q}^n$, and $\tau \in \mathbb{Q}$, the source question is whether there is an $x \in \mathbb{R}^n$ such that

$$Ax \leq b, \quad c^\top x \geq \tau.$$

The reduction maps this question to whether the optimum of a positive packing linear program reaches a specified threshold. It first computes a box of polynomial encoding length that contains a feasible point whenever one exists. The output has at most $2n$ variables and at most $m + n + 1$ constraints, and its constraint matrix P satisfies $\text{nnz}(P) \leq \text{nnz}(A) + \text{nnz}(c) + 2n$. Every output coefficient has polynomial encoding length. Once the rational rows have been normalized and the box computed, the sparse core uses $O(\text{nnz}(A) + \text{nnz}(c) + m + n)$ exact arithmetic operations and output writes. An explicit packing point at the threshold decodes to a source feasible point in $O(n)$ time.

The reduction writes the objective threshold as one additional linear inequality, splits signed variables into nonnegative pairs, shifts each signed row using the computed box bound, and uses the positive objective to force the bounding inequalities to be tight. One motivation is the quantitative reduction of Kyng, Wang, and Zhang linking accurate packing-LP solving to linear-system solving. The present construction is deliberately exact rather than approximation-preserving: infeasible source instances can map to packing optima arbitrarily close to the threshold. It therefore does not by itself transfer running times for multiplicative positive-LP approximation.

Note. This paper was fully AI generated with ChatGPT 5.5 Pro from prompts by Xinyuan Cao, Hanxi Lin, and Richard Peng. The present repository version, including this disclosure note, was audited and revised by GPT-5 in the Codex harness.

1 Introduction

General linear programs may have signed variables and signed constraint, objective, and right-hand-side data. Positive linear programming conventionally refers to the packing/covering pair with nonnegative data [LN93]. We use the packing form, which maximizes a nonnegative

objective subject to nonnegative upper-bound constraints, and give an exact sparse conversion from a general LP threshold decision to a positive packing threshold.

Let $A \in \mathbb{Q}^{m \times n}$, $b \in \mathbb{Q}^m$, $c \in \mathbb{Q}^n$, and $\tau \in \mathbb{Q}$. The source question is

$$\exists x \in \mathbb{R}^n \quad Ax \leq b, \quad c^\top x \geq \tau. \quad (1)$$

Appending the inequality $-c^\top x \leq -\tau$ turns this question into rational linear feasibility. A polynomial-bit box containing a feasible point can be computed directly from the rational input. The algebraic core of the reduction therefore considers arbitrary real $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ together with $U > 0$, and handles

$$Ax \leq b, \quad -U\mathbf{1} \leq x \leq U\mathbf{1}. \quad (2)$$

The entries of A and b may have either sign. The target is a positive packing linear program

$$\max\{e^\top z : z \geq 0, \quad Pz \leq d\}, \quad (3)$$

in which P , d , and e are entrywise nonnegative. Our goal is not to solve either program, but to convert (2) to an exact objective-threshold question for (3) while preserving sparsity. Combining this core with the one-row objective reduction and the polynomial-bit box gives the claimed reduction for the general rational question (1).

For rational input, general linear programming is polynomial-time solvable by the ellipsoid and interior-point frameworks [Kha79, Kar84]. That bit-complexity statement is distinct from the unit-cost construction bound proved here. Exact positive linear programming also has a parallel-complexity barrier: Trevisan and Xhafa proved that exact positive-LP solution is P-complete, so an NC algorithm would imply $P = NC$ [TX98].

Much of the algorithmic literature therefore concerns approximation. For a packing program with optimum OPT , a $(1 - \varepsilon)$ -approximate solution is a feasible point of value at least $(1 - \varepsilon)\text{OPT}$. Luby and Nisan initiated width-independent parallel approximation for positive linear programs [LN93]; Plotkin, Shmoys, and Tardos developed the Lagrangian-relaxation framework for fractional packing and covering [PST95]. For an explicit constraint matrix with N nonzeros, r rows, and s columns, Koufogiannakis and Young gave, with high probability, feasible primal and dual solutions within a factor $1 + \varepsilon$ of optimum in

$$O\left(N + \frac{(r + s) \log N}{\varepsilon^2}\right)$$

time [KY14]. Allen-Zhu and Orecchia subsequently obtained $\tilde{O}(N/\varepsilon)$ time for packing and $\tilde{O}(N/\varepsilon^{3/2})$ time for covering [AZO19].

The direct motivation for asking about this conversion comes from Kyng, Wang, and Zhang [KWZ20]. They give quantitative reductions from conditioned least-squares problems to packing programs, showing that sufficiently fast and accurate packing solvers would yield faster algorithms for linear systems. More precisely, for a full-column-rank matrix $A \in \mathbb{R}^{m \times n}$ and a known bound $K_A \geq \kappa(A)$, their sparse reduction turns a $\tilde{O}(N/\varepsilon^\alpha)$ -time packing solver into a constant-relative-residual least-squares solver running in $\tilde{O}(\text{nnz}(A)(m^{3/2}K_A^3)^\alpha)$ time. The condition-number bound determines the solution scale and packing accuracy used for residual decoding and iterative refinement.

At the center of our reduction is a simpler exact conversion from a signed system inside a known box to positive linear programming. For rational input, a determinant bound supplies a suitable box of polynomial encoding length, so no boundedness promise is imposed on the source feasibility problem. At a structural level, our construction and that of Kyng, Wang, and Zhang both split signed variables and use an objective to force bounding constraints to be tight. The guarantee here is otherwise different: it does not provide the quantitative approximation-to-residual implication needed for the linear-system application. We make no novelty claim for the shared algebraic devices. The purpose of this note is to give a self-contained exact formulation, exact sparse accounting, and an explicit statement of what near-threshold points do and do not imply.

We first state the boxed-feasibility primitive.

Theorem 1.1 (Boxed linear feasibility to positive linear programming). *Let m, n be positive integers, let $A \in \mathbb{R}^{m \times n}$ be given in sparse row form, let $b \in \mathbb{R}^m$, and let $U > 0$. In the exact sparse arithmetic model, a deterministic algorithm constructs, using $O(\text{nnz}(A) + m + n)$ arithmetic operations and output writes, a packing linear program*

$$\max\{e^\top z : z \in \mathbb{R}_{\geq 0}^q, Pz \leq d\}$$

and a threshold $\theta > 0$ such that P , d , and e are entrywise nonnegative,

$$q \leq 2n, \quad \text{nnz}(P) \leq \text{nnz}(A) + 2n,$$

and

$$\exists x \in [-U, U]^n \text{ with } Ax \leq b \iff \max\{e^\top z : z \geq 0, Pz \leq d\} \geq \theta.$$

In the nontrivial branch the output has $q = 2n$, exactly $m + n$ constraints, $\theta = nU$, and $\text{nnz}(P) = \text{nnz}(A) + 2n$. A boxed feasible point can be encoded as a packing point of value θ in $O(n)$ operations; conversely, any explicit feasible packing point of value at least θ decodes to a boxed feasible point in $O(n)$ operations.

Lemma 1.2 (A polynomial-bit box for rational feasibility). *Let p, n be positive integers, let $B \in \mathbb{Q}^{p \times n}$, and let $d \in \mathbb{Q}^p$. Write the entries in lowest terms with positive denominators. For each row i , let q_i be the product of the denominators appearing in that row, including the denominator of d_i , and multiply the row by q_i . This gives an equivalent integral system $\bar{B}x \leq \bar{d}$. Define*

$$M = \max(\{1\} \cup \{|\bar{B}_{ij}| : i \in [p], j \in [n]\} \cup \{|\bar{d}_i| : i \in [p]\}), \quad r = \min\{p, n\},$$

and $U = (rM)^r$. Then

$$\{x \in \mathbb{R}^n : Bx \leq d\} \neq \emptyset \iff \{x \in [-U, U]^n : \bar{B}x \leq \bar{d}\} \neq \emptyset.$$

Moreover,

$$\log_2 U \leq r(\log_2 r + \log_2 M),$$

so U has polynomial encoding length in the binary encoding of (B, d) .

Proof. Only the forward implication requires proof. Write $x = x^+ - x^-$ and introduce slacks:

$$\overline{B}x^+ - \overline{B}x^- + s = \overline{d}, \quad x^+, x^-, s \geq 0.$$

If this standard-form system is feasible, it has a basic feasible solution. One way to see this is to minimize the sum of all variables: the sublevel set below the value of any fixed feasible point is compact, so an optimum exists, and an extreme optimum is basic.

Let C be the integral basis matrix of such a solution, and suppose that k of its columns come from $(\overline{B}, -\overline{B})$ rather than from the slack identity matrix. A nonsingular basis cannot contain both $\overline{B}_{*,j}$ and $-\overline{B}_{*,j}$, so $k \leq \min\{p, n\} = r$. Expanding along the $p - k$ slack columns reduces $\det C$ to a $k \times k$ integral determinant. Hence $|\det C| \geq 1$. For any basic coordinate belonging to x^+ or x^- , Cramer's rule and the same expansion bound the numerator by

$$k!M^k \leq (rM)^r = U.$$

At most one of x_j^+ and x_j^- is basic; every nonbasic variable is zero. Therefore $|x_j| \leq U$ for every j . The binary length of q_i is at most the sum of the binary lengths of the denominators in row i . Hence the normalized integer data, M , and U all have encoding length polynomial in that of (B, d) . $\square$

Corollary 1.3 (Rational LP threshold decision to a positive packing threshold). *Let $A \in \mathbb{Q}^{m \times n}$, $b \in \mathbb{Q}^m$, $c \in \mathbb{Q}^n$, and $\tau \in \mathbb{Q}$ be binary encoded, with m, n positive. A deterministic polynomial-time algorithm constructs a positive packing linear program and a threshold $\theta > 0$ such that*

$$\exists x \in \mathbb{R}^n \quad Ax \leq b, \quad c^\top x \geq \tau \quad \Longleftrightarrow \quad \text{OPT}_{\text{pack}} \geq \theta.$$

The output has at most $2n$ variables and at most $m + n + 1$ constraints, its constraint matrix P satisfies

$$\text{nnz}(P) \leq \text{nnz}(A) + \text{nnz}(c) + 2n,$$

and every output coefficient has encoding length polynomial in the input encoding length. An explicit packing point of value at least θ decodes to a source feasible point in $O(n)$ exact arithmetic operations.

Proof. Append the objective row and write the resulting feasibility system as

$$Bx \leq d, \quad B = \begin{pmatrix} A \\ -c^\top \end{pmatrix}, \quad d = \begin{pmatrix} b \\ -\tau \end{pmatrix}.$$

Normalize each row using the denominator product specified in Lemma 1.2, and compute U from that lemma. Applying Theorem 1.1 to the normalized system inside $[-U, U]^n$ proves the equivalence and the dimension, sparsity, and witness claims. Row normalization preserves the zero pattern. Its coefficient lengths are polynomial in the input length, as is $\log U$; the shifted coefficients, shifted right-hand sides, box bounds, and threshold therefore also have polynomial encoding length. All normalization, integer arithmetic, and sparse-output construction can be performed in polynomial bit time. $\square$

Proof idea. For rational input, first write the objective threshold as one additional row, clear denominators, and use Lemma 1.2 to compute a polynomial-bit coordinate bound. For the sparse boxed-feasibility core, write $x_j = y_j - s_j$ with $y_j, s_j \geq 0$ and intend to enforce $y_j + s_j = U$. For row i , put

$$R_i = \sum_j |A_{ij}|, \quad p_{ij}^+ = A_{ij} + |A_{ij}|, \quad p_{ij}^- = -A_{ij} + |A_{ij}|.$$

The shifted row

$$\sum_j p_{ij}^+ y_j + \sum_j p_{ij}^- s_j \leq b_i + U R_i$$

has nonnegative coefficients and, on the intended slice, is exactly $A_i x \leq b_i$. The packing constraints $y_j + s_j \leq U$ and the objective $\sum_j (y_j + s_j)$ with threshold nU force every one of these inequalities to be tight. If some shifted right-hand side is negative, the corresponding original row is infeasible throughout the box and the algorithm returns a fixed rejecting instance.

Scope. Removing the supplied box uses the finite binary encoding of rational data. For arbitrary exact-real data, Theorem 1.1 still requires U as input. The computed U may have exponential magnitude; only its encoding length is polynomial. Thus Corollary 1.3 is a polynomial bit-time reduction, not a strongly polynomial result. The $O(\text{nnz}(A) + m + n)$ bound in Theorem 1.1 counts exact unit-cost arithmetic operations and output writes after U is available, and it does not include the cost of solving the output LP. The construction is an exact-threshold reduction, not a promise-gap reduction. We prove in Section 4 that near-threshold packing points decode to additively near-feasible boxed points, but also that infeasible instances can have packing optimum arbitrarily close to nU . Thus a value-only threshold oracle does not supply a witness, and a standard multiplicative approximation does not decide arbitrary boxed feasibility.

The remainder of the paper is organized as follows. Section 2 records the algebraic identity behind the shift. Section 3 gives the algorithm and proves the feasibility primitive. Section 4 gives the approximate decoding bound and the no-gap example.

2 Setup

All vector inequalities are coordinatewise. For a positive integer k , write $[k] = \{1, \dots, k\}$, and let $\mathbf{1}$ denote an all-ones vector of the dimension determined by context. Empty sums are zero.

Definition 2.1 (Positive packing threshold instance). *A packing linear program, the maximization form of a positive linear program, has the form*

$$\max\{e^\top z : z \in \mathbb{R}_{\geq 0}^q, Pz \leq d\},$$

where P , d , and e are entrywise nonnegative. A packing threshold instance also specifies $\theta > 0$ and accepts if the maximum is at least θ .

Sparse arithmetic model. The input matrix is available in sparse row form. Reading and writing a stored coefficient, adding or comparing real numbers, taking an absolute value, and multiplying by U each cost one operation. These are exact operations. The cost of writing the sparse output and its dense right-hand side and objective vectors is included.

For $i \in [m]$ and $j \in [n]$, define

$$R_i = \sum_{j=1}^n |A_{ij}|, \quad p_{ij}^+ = A_{ij} + |A_{ij}|, \quad p_{ij}^- = -A_{ij} + |A_{ij}|, \quad c_i = b_i + UR_i. \quad (4)$$

The quantities R_i, p_{ij}^+, p_{ij}^- are nonnegative. The right-hand side c_i need not be.

Observation 2.2 (Shift identity and tight slice). *For all $y, s \in \mathbb{R}_{\geq 0}^n$ and $i \in [m]$,*

$$\sum_{j=1}^n p_{ij}^+ y_j + \sum_{j=1}^n p_{ij}^- s_j = \sum_{j=1}^n A_{ij}(y_j - s_j) + \sum_{j=1}^n |A_{ij}|(y_j + s_j). \quad (5)$$

If $y_j + s_j \leq U$ for every j and $\sum_j (y_j + s_j) \geq nU$, then $y_j + s_j = U$ for every j .

Proof. Expanding the definitions of p_{ij}^+ and p_{ij}^- gives (5). For the second assertion, set $\delta_j = U - y_j - s_j \geq 0$. The objective inequality says $\sum_j \delta_j \leq 0$, so every nonnegative δ_j is zero. $\square$

The tight slice is in bijection with the box. If $x \in [-U, U]^n$, then

$$y_j = \frac{U + x_j}{2}, \quad s_j = \frac{U - x_j}{2} \quad (6)$$

is nonnegative, $y_j + s_j = U$, and $y_j - s_j = x_j$. Conversely, nonnegative y_j, s_j with sum U give $x_j = y_j - s_j \in [-U, U]$.

3 The reduction to positive linear programming

We first describe the algorithm. Coefficients equal to zero are not stored.

Algorithm BoxToPACKING. Given A, b, U , perform the following steps.

1. Compute R_i and c_i from (4) for every $i \in [m]$.
2. If $c_i < 0$ for some i , return the one-variable packing program

$$\max\{0 \cdot z : z \geq 0, z \leq 0\}$$

with threshold $\theta = 1$.

3. Otherwise, use variables

$$z = (y_1, \dots, y_n, s_1, \dots, s_n) \in \mathbb{R}_{\geq 0}^{2n},$$

set the objective vector to the all-ones vector, and set $\theta = nU$.

4. For each $i \in [m]$, create the shifted row

$$\sum_{j=1}^n p_{ij}^+ y_j + \sum_{j=1}^n p_{ij}^- s_j \leq c_i. \quad (7)$$

5. For every $j \in [n]$, create the box row

$$y_j + s_j \leq U. \quad (8)$$

Return these $m + n$ constraints, the objective, and the threshold.

The exceptional branch is justified by the following elementary certificate.

Lemma 3.1 (A negative shifted right-hand side certifies infeasibility). *If $c_i < 0$ for some $i \in [m]$, then the boxed system is infeasible.*

Proof. For every $x \in [-U, U]^n$,

$$(Ax)_i \geq -\sum_{j=1}^n |A_{ij}| |x_j| \geq -UR_i.$$

The inequality $c_i = b_i + UR_i < 0$ is equivalent to $b_i < -UR_i$. Hence $(Ax)_i > b_i$ throughout the box. $\square$

We next prove the exact equivalence in the main branch.

Lemma 3.2 (Threshold equivalence and witness conversion). *Assume $c_i \geq 0$ for every $i \in [m]$. The boxed system is feasible if and only if the packing program defined by (7) and (8) has optimum at least nU . The maps $x \mapsto (y, s)$ from (6) and $(y, s) \mapsto y - s$ convert explicit witnesses in $O(n)$ operations.*

Proof. Suppose $x \in [-U, U]^n$ and $Ax \leq b$. Define y, s by (6). They satisfy every box row with equality. By (5), for each $i \in [m]$,

$$\sum_j p_{ij}^+ y_j + \sum_j p_{ij}^- s_j = (Ax)_i + UR_i \leq b_i + UR_i = c_i.$$

Thus (y, s) is packing-feasible and has objective nU .

Conversely, the origin is packing-feasible because every c_i is nonnegative. The box rows make the feasible region compact, so the maximum is attained. Let (y, s) be feasible with objective at least nU . The box rows also bound that objective by nU ; hence Observation 2.2 gives $y_j + s_j = U$ for every j . Put $x = y - s$. Then $x \in [-U, U]^n$, and (5) gives

$$(Ax)_i + UR_i = \sum_j p_{ij}^+ y_j + \sum_j p_{ij}^- s_j \leq c_i = b_i + UR_i.$$

Therefore $Ax \leq b$. Both conversions require constant work per coordinate. $\square$

Proof of Theorem 1.1. Run BOXTOPACKING. If the exceptional branch is taken, its packing optimum is $0 < 1 = \theta$, while Lemma 3.1 says that the boxed system is infeasible. The branch has one variable, one constraint, and one matrix nonzero; because $n \geq 1$,

$$1 \leq \text{nnz}(A) + 2n.$$

In the main branch, every coefficient in (7) is nonnegative, every c_i is nonnegative, and the box rows, objective, and threshold are positive or nonnegative as required. Thus the output is a packing threshold instance. Lemma 3.2 proves the exact equivalence and the witness statements.

It remains to count the output and the construction work. If $A_{ij} > 0$, then $p_{ij}^+ = 2A_{ij}$ and $p_{ij}^- = 0$; if $A_{ij} < 0$, then $p_{ij}^+ = 0$ and $p_{ij}^- = -2A_{ij}$. Consequently, each stored nonzero of A produces exactly one nonzero in the shifted rows. The n box rows contribute two nonzeros each, so

$$\text{nnz}(P) = \text{nnz}(A) + 2n$$

in the main branch. A constant number of sparse scans computes the row radii and writes the shifted coefficients. Initializing the m right-hand sides, the $2n$ objective entries, and the n box rows costs $O(m+n)$ more operations and writes. This proves the claimed $O(\text{nnz}(A) + m + n)$ bound. $\square$

4 The approximation boundary

Although the exact reduction has no promise gap, its feasible points have a simple quantitative interpretation below the threshold.

Proposition 4.1 (Decoding a near-threshold point). *Consider a main-branch instance and any packing-feasible point (y, s) . Define*

$$x = y - s, \quad \delta_j = U - y_j - s_j, \quad \Delta = \sum_{j=1}^n \delta_j = nU - \sum_{j=1}^n (y_j + s_j).$$

Then $x \in [-U, U]^n$, and for every $i \in [m]$,

$$(Ax)_i \leq b_i + \sum_{j=1}^n |A_{ij}| \delta_j \leq b_i + \|A_{i,*}\|_\infty \Delta. \quad (9)$$

In particular, objective value at least $(1 - \varepsilon)nU$ implies

$$(Ax)_i \leq b_i + \varepsilon nU \|A_{i,*}\|_\infty \quad (i \in [m]).$$

Proof. Nonnegativity and the box rows give $|x_j| = |y_j - s_j| \leq y_j + s_j \leq U$. Rearranging the shifted-row inequality with (5) yields

$$(Ax)_i \leq b_i + UR_i - \sum_j |A_{ij}|(y_j + s_j) = b_i + \sum_j |A_{ij}| \delta_j.$$

The second inequality in (9) follows from $\delta_j \geq 0$ and $\sum_j \delta_j = \Delta$. The final statement uses $\Delta \leq \varepsilon nU$. $\square$

The additive bound in Proposition 4.1 does not yield an exact feasibility decision without a separate promise. The next example shows that this is unavoidable for the present construction.

Proposition 4.2 (No uniform objective gap). *For every $\delta \in (0, 1)$, there is an infeasible boxed instance with $m = 2$, $n = 1$, and $U = 1$ whose output takes the main branch and has*

$$\text{OPT} = \theta - \frac{\delta}{2}.$$

Consequently, the separation between infeasible and feasible outputs can be arbitrarily small, both additively and multiplicatively.

Proof. Take

$$A = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ -\delta \end{pmatrix}, \quad U = 1.$$

The boxed system requires $x \leq 0$ and $x \geq \delta$, so it is infeasible. Both row radii are 1, and the shifted right-hand sides are 1 and $1 - \delta$, so the main branch is taken. Its variables are $y, s \geq 0$, its threshold is $\theta = 1$, and its constraints are

$$2y \leq 1, \quad 2s \leq 1 - \delta, \quad y + s \leq 1.$$

The first two bounds are simultaneously attainable and imply

$$\text{OPT} = \frac{1}{2} + \frac{1 - \delta}{2} = 1 - \frac{\delta}{2}.$$

Letting δ tend to zero proves the claim. □

Thus the exact threshold equivalence alone cannot support an approximation-preserving reduction. Any such extension would need an additional promise or a different construction that creates and tracks a quantitative gap.

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